DNA Repair

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Causes of DNA damage

- *Exogenous* : caused by UV lights, X ray, gamma rays, herbicides insectisides,viruses, alkylating agents and intercalating agents, pollution, smoking…

- *Endogenous*: due to replication errors, oxidants and ROS

Mutation:

A permanent change in the nucleotide sequence. This may affect :

1- natures of the bases :

A- Substitution mutation:

Replacement of one/more base pair with another.

B- Insertion or deletion mutation:

addition or deletion of one/more base pairs.

2- effect on the coding sequence and the product

A- Silent mutation:

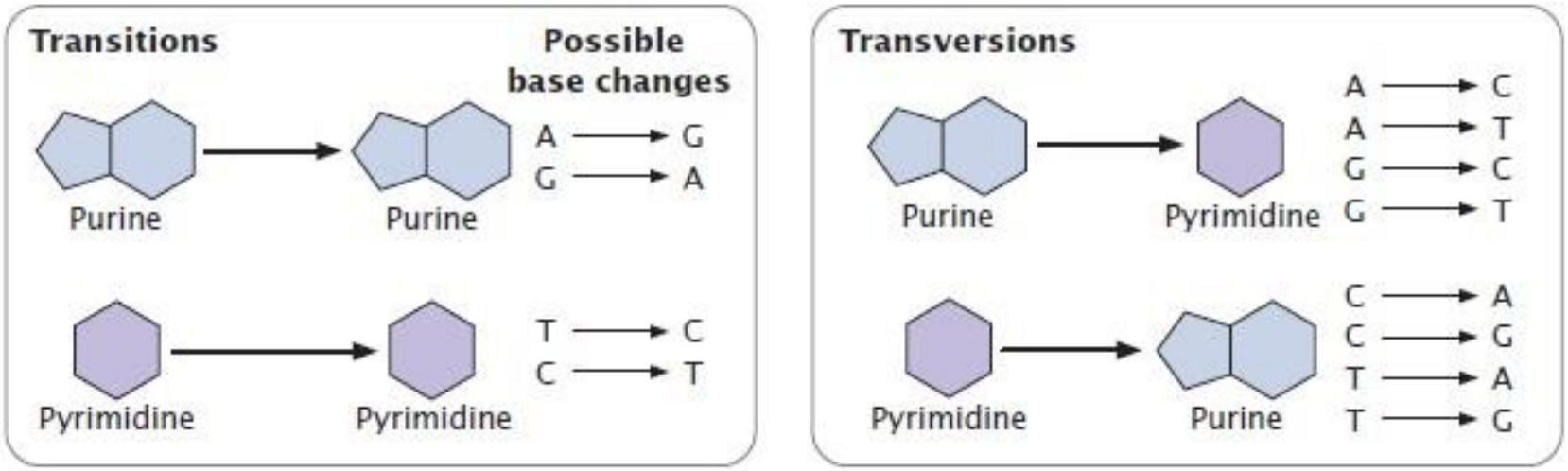
mutation affects non essential DNA, or has neglieable effect.

B-nonsense mutation: premature stop codon

C- missense mutation: change in amino acid encoded

Base substitution:

Transitions Vs transversions



18.3 A transition is the substitution of a purine for a purine or of a pyrimidine for a pyrimidine; a transversion is the substitution of a pyrimidine for a purine or of a purine for a pyrimidine.

Mutations and Cancer

Strong correlation between carcinogens & Mutagens.

90% of carcinogens are mutagens.

Ames Test for Carcinogens:

Measure the potential of a chemical to induce mutations in bacteria.

This means it can be a carcinogen.

Rapid and inexpensive Screen important in industry:

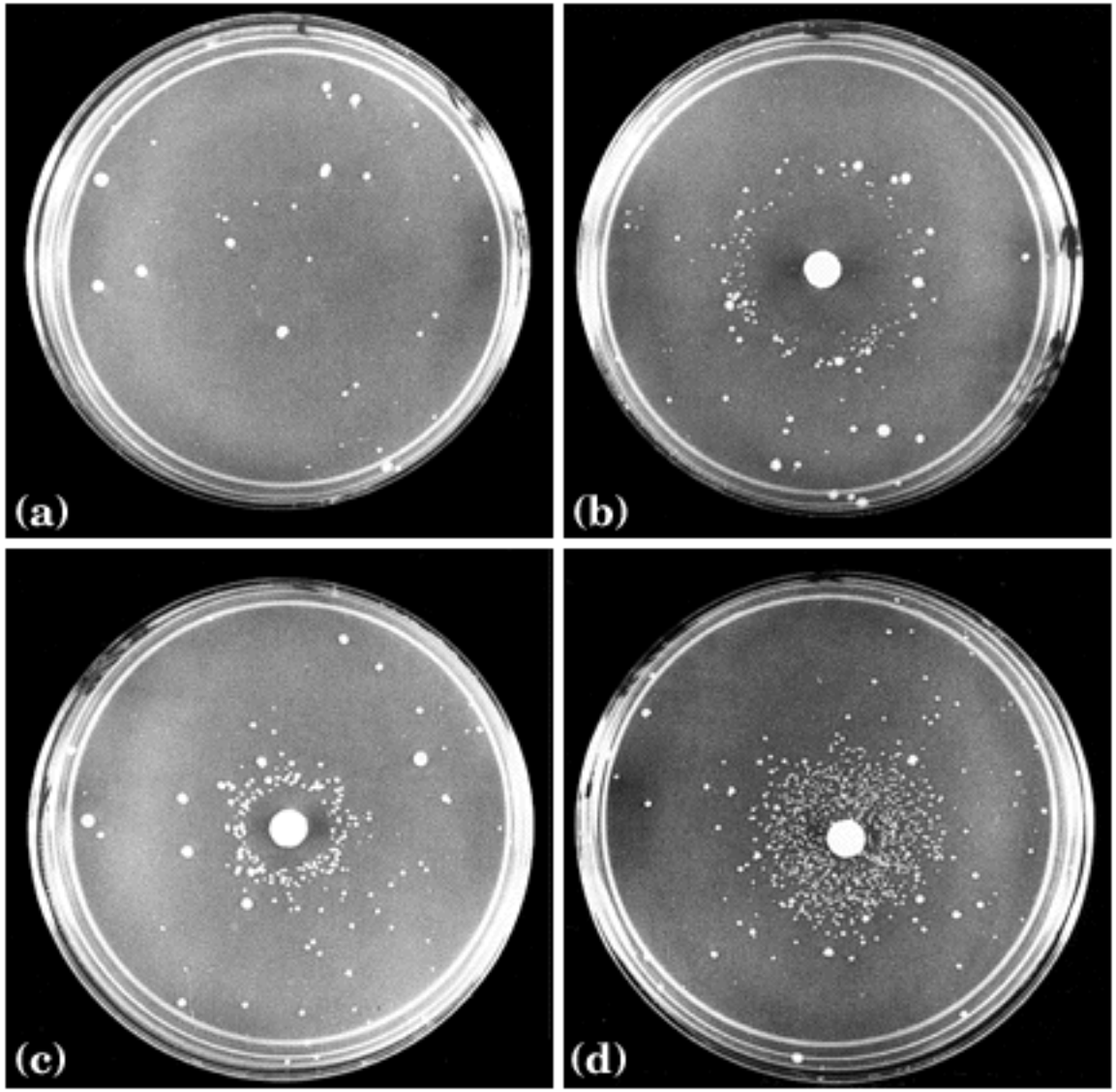
Salmonella carry mutations in genes involved in histidine synthesis (they require histidine for growth).

Bacteria spread on an agar plate with a small amount of histidine (to allow bacteria to grow for an initial time and have the opportunity to mutate).

When histidine depleted →only mutated bacteria that can produce its own histidine will survive.

Plate incubated for 48h.

Mutagenicity of a substance is proportional to the # of colonies observed.



Number and diversity of DNA repair pathways reflects

1) various sources of DNA damage

2) importance of repair pathways.

- DNA repair inefficient energetically in comparison to metabolism (ATP accounted for and used optimally → integrity of the genetic material is of greatest value.

- DNA damage of one strand removed / corrected using undamaged strand as a template.

table 25–5

Types of DNA Repair Systems in *E. coli*

Enzymes/proteins	Type of damage
Mismatch repair	
Dam methylase	Mismatches
MutH, MutL, MutS proteins	
DNA helicase II	
SSB	
DNA polymerase III	
Exonuclease I	
Exonuclease VII	
RecJ nuclease	
Exonuclease X	
DNA ligase	Deamination of C deamination of A
Base-excision repair	
DNA glycosylases	Abnormal bases (uracil, hypoxanthine, xanthine); alkylated bases; pyrimidine dimers in some other organisms *
AP endonucleases	Methylated A/G
DNA polymerase I	
DNA ligase	
Nucleotide-excision repair	
ABC excinuclease	DNA lesions that cause large structural changes (e.g., pyrimidine dimers) *
DNA polymerase I	
DNA ligase	
Direct repair	
DNA photolyases	Pyrimidine dimers*
O^6-Methylguanine-DNA methyltransferase	O^6-Methylguanine

1) Mismatch Repair:

Improves fidelity by a factor of 10^2-10^3.

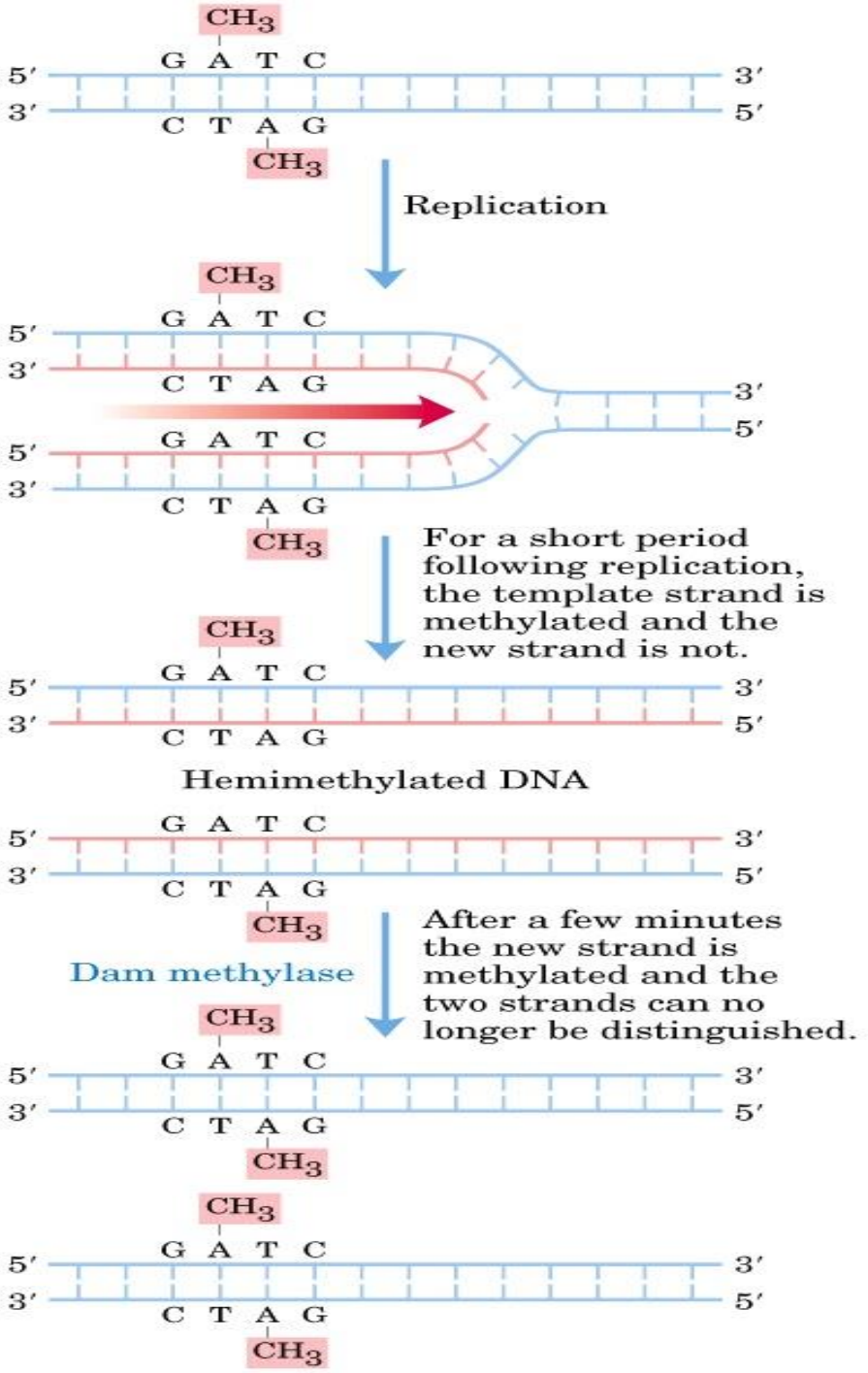
Discrimination between template and newly synthesized strand by tagging the template with methyl groups.

A short period where template is distinguished from newly synthesized

Repair occurs in this period→→→

lag period = 10min.

Dam methylase 5'-GATC-3' in newly synthesized. So both strands are methylated and indistinguishable.



Methyl- directed mismatch repair:

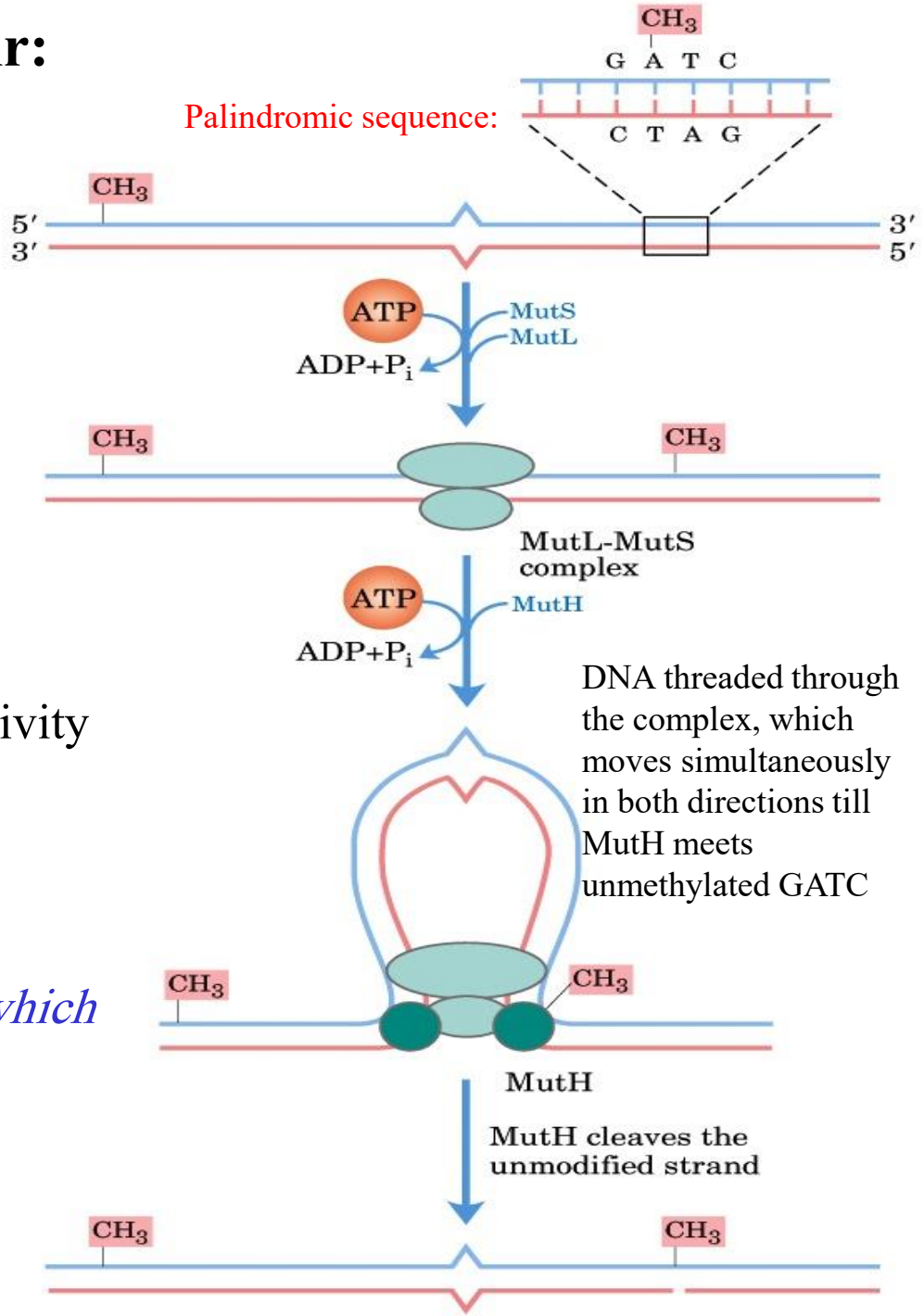
Repairs up to 1000bp from hemimethylated GATC sequence.

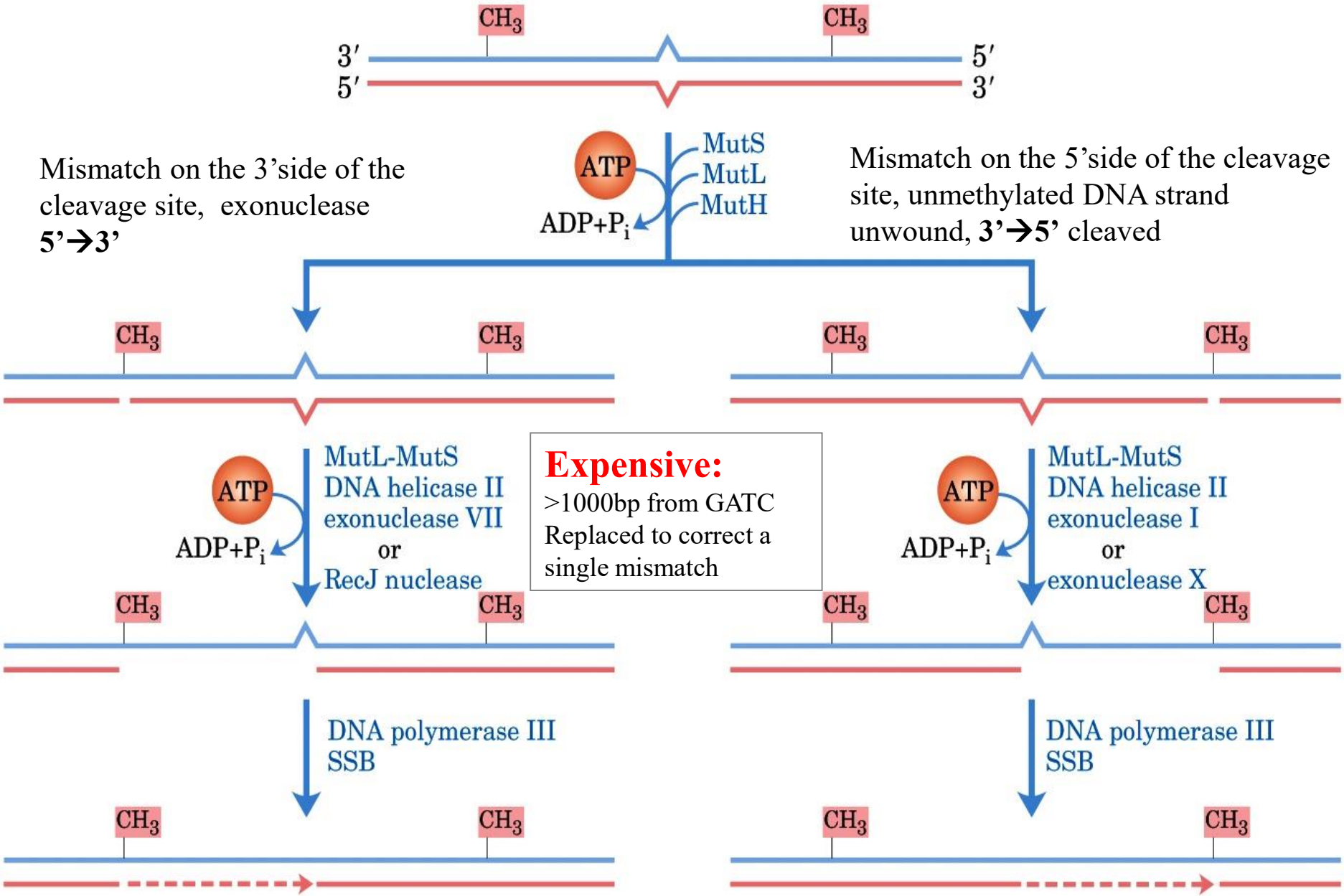
MutS +MutL protein repair complex binds all mismatched base pair.

MutH binds MutL.

MutH has a site specific endonuclease activity inactive till meets unmethylated 5'-GATC-3' (marks the strand for repair).

Human homologs mutated in CANCER which increases the mutation rate



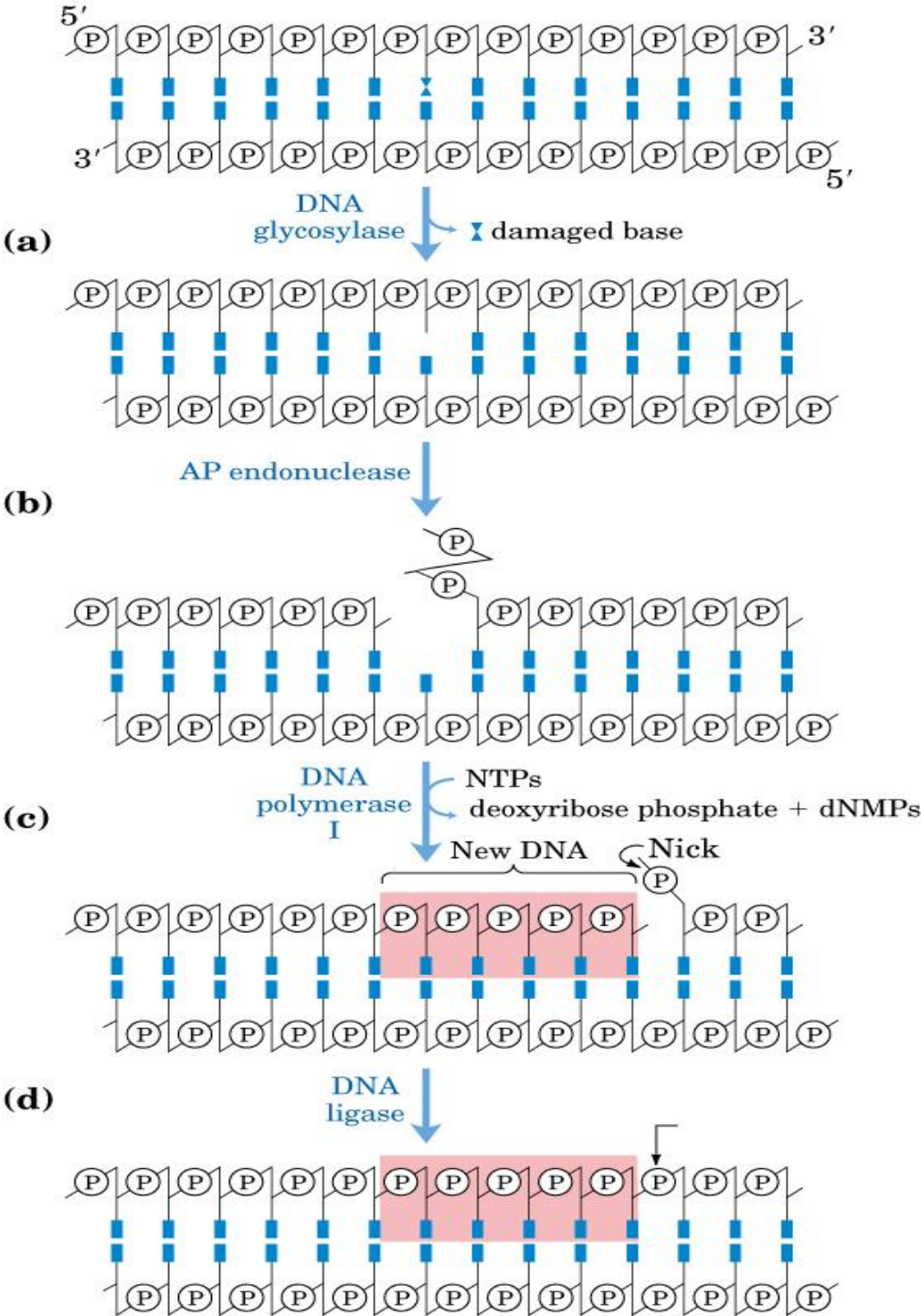

CH3
CH3
3′
5′
5′
3′
ATP
ADP+Pi
MutS
MutL
MutH
Mismatch on the 3'side of the cleavage site, exonuclease 5'→3'
Mismatch on the 5'side of the cleavage site, unmethylated DNA strand unwound, 3'→5' cleaved
MutL-MutS
DNA helicase II
exonuclease VII
or
RecJ nuclease
MutL-MutS
DNA helicase II
exonuclease I
or
exonuclease X
Expensive:
>1000bp from GATC
Replaced to correct a single mismatch
DNA polymerase III
SSB

2) Base- Excision Repair:

DNA glycosylases recognize DNA lesions (products of A/C deamination. Caused by *environmental agents.*
Remove affected base by cleaving N-glycosyl bond bw base & deoxyribose → apurinic/apyrimidinic base = Abasic site= AP site.

Once an AP site has formed→→ must repair.
AP endonuclease (cut DNA strand containing AP site)

DNA segment containing the AP site is removed → DNA pol I replaces DNA, ligase seals the nick.



- Uracil DNA glycosylase (UNG)→ removes U formed from deamination of C.

- Highly selective distinguish T from U (act only on DNA not RNA)

- Bacteria has one type, humans 4.

3) Nucleotide Excision Repair: (critical for all organisms survival)

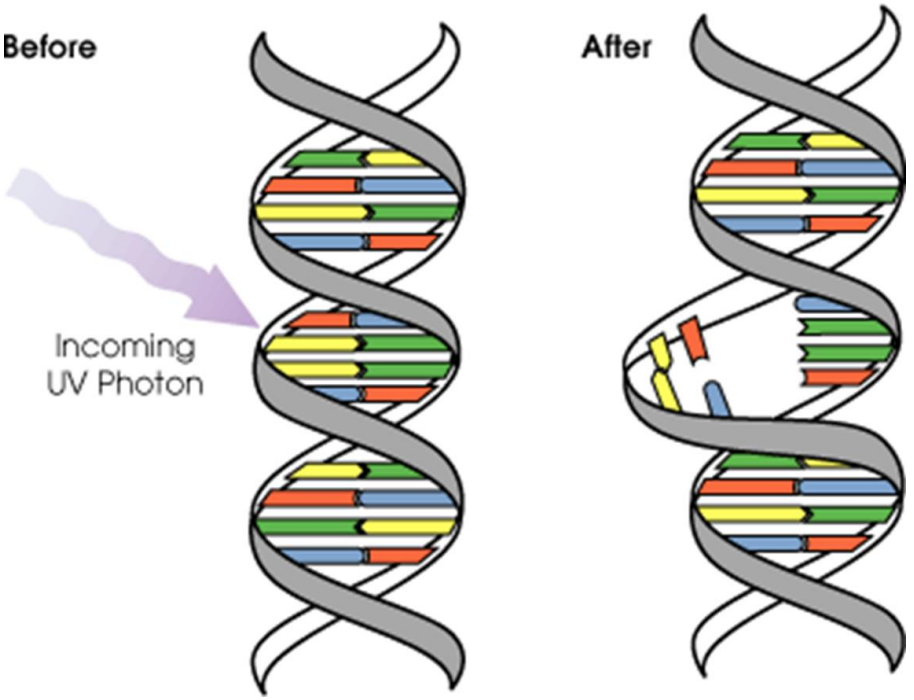
Recognize bulky lesions and pyrimidine dimers).

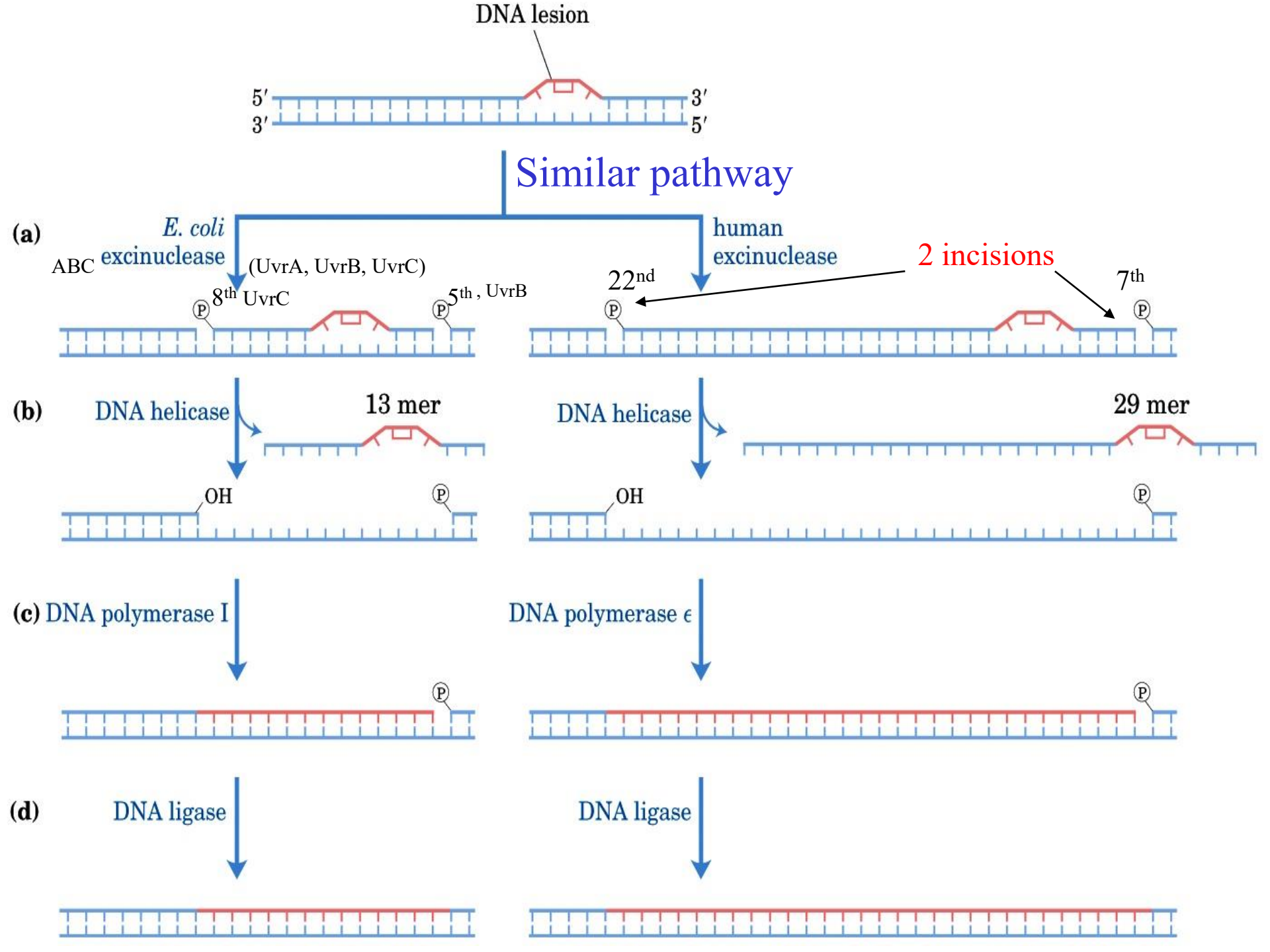
DNA lesions cause large distortion in DNA helical structure.

Multisubunit enzyme hydrolyses 2 phosphodiester bonds one on either side of the distorsion.

Excinuclease= catalyze 2 specific endonucleolytic cleavages. (not standard endonuclease)

Thymine dimer




DNA lesion
5′
3′
3′
5′
Similar pathway
(a)
E. coli
ABC excinuclease
(UvrA, UvrB, UvrC)
8th UvrC
5th , UvrB
human
excinuclease
2 incisions
22nd
7th
(b)
DNA helicase
13 mer
OH
DNA helicase
29 mer
OH
(c) DNA polymerase I
DNA polymerase ϵ
(d)
DNA ligase
DNA ligase

This pathway the primary route for many lesion types :

1- pyrimidine dimers (T, C).

2- benzo pyrene-guanine (formed in DNA by **cigarette smoke**).

Xeroderma pigmentosum (XP):

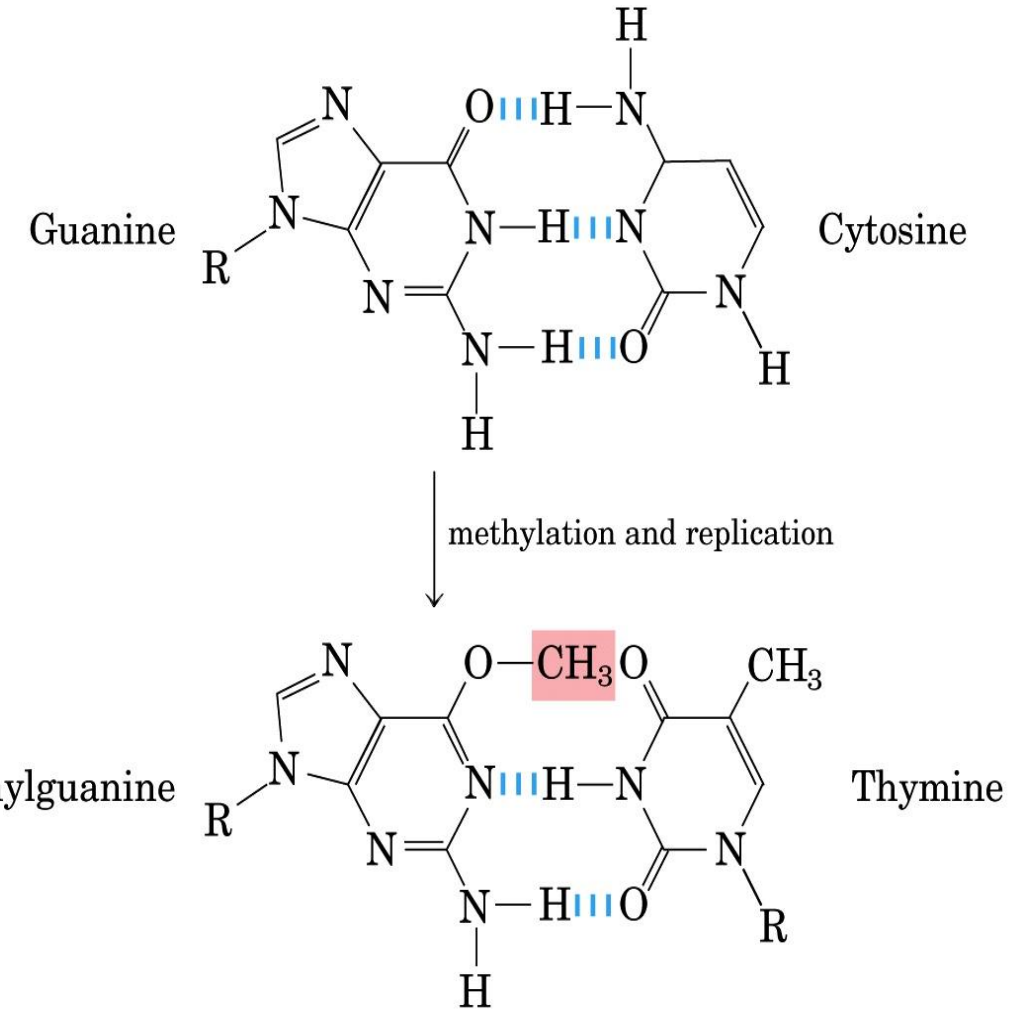
rare inherited disease (pigmented lesions on skin + skin cancer) due to mutations in NER system.

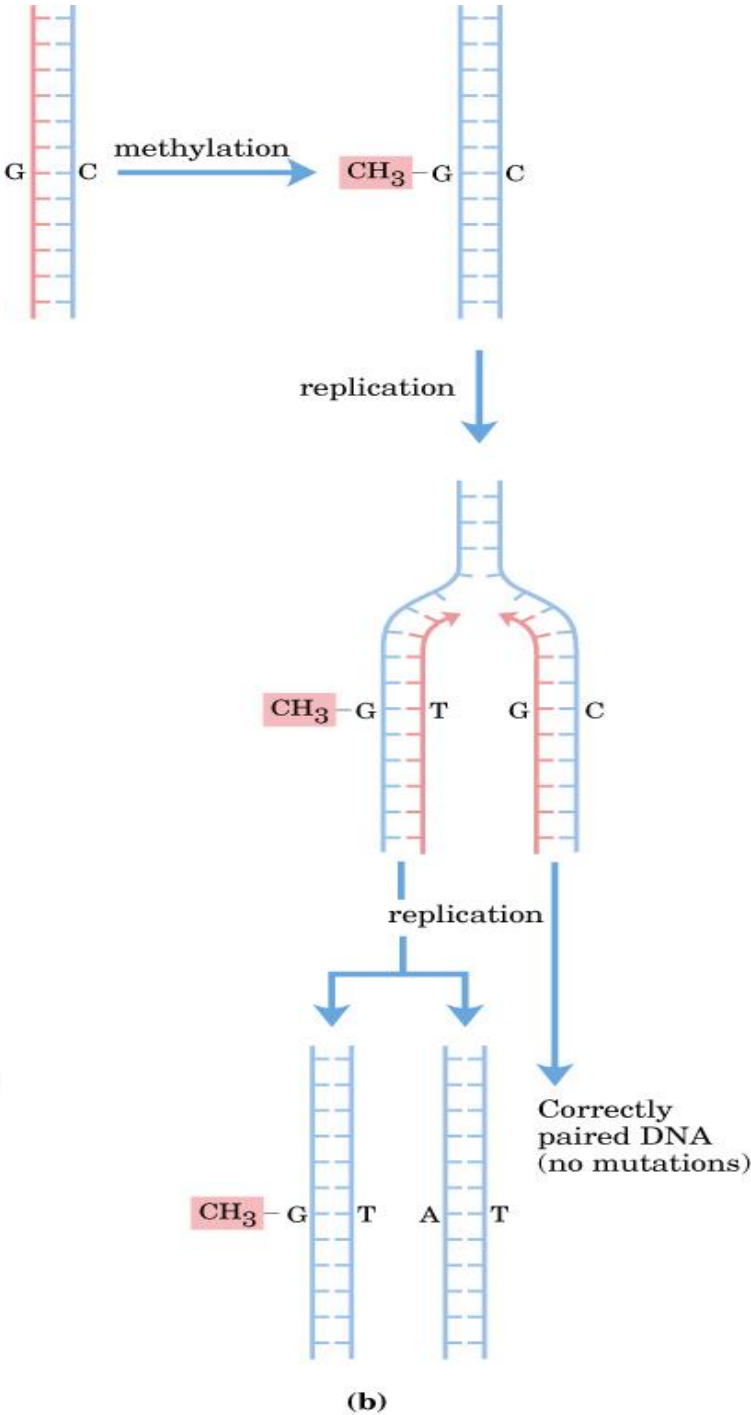
Hereditary non-polyposis colorectal cancer (HNPCC) :

Inherited disease with defects in mismatch repair. Patients suffer from rectal bleeding, stomach pain , weight loss. Most commom mutation in hMLH1(human Mutl homolog1) & hMSH2 (human Muts homolog2)

DNA damage cause mutations:

The methylation product of alkylating agents pairs with T rather than C during replication.





b-Direct repair :

Repair with no excision / removal of a base / a nucleotide.

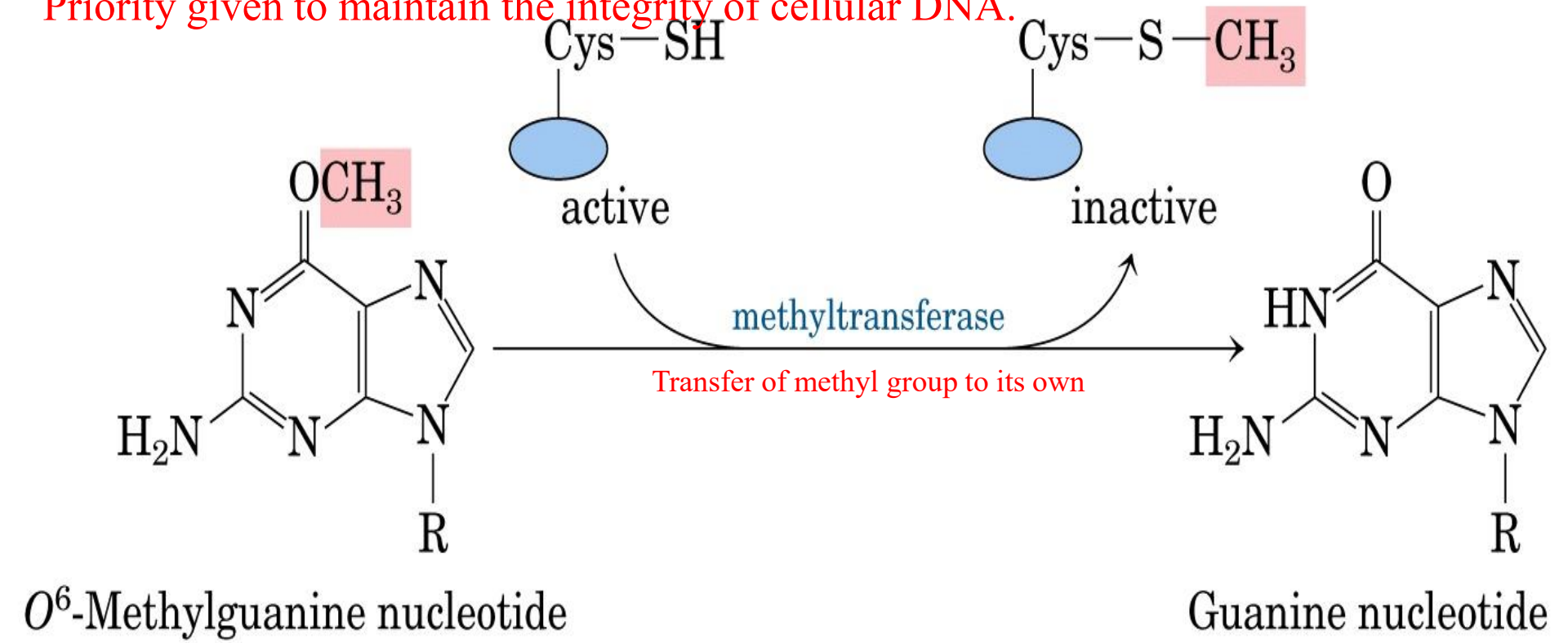
O_6- methylguanine- DNA methyltransferase.

Not strictly an enzyme.

A single methyl transfer event permanently methylates the protein.

An entire protein consumed to correct a single damaged base →

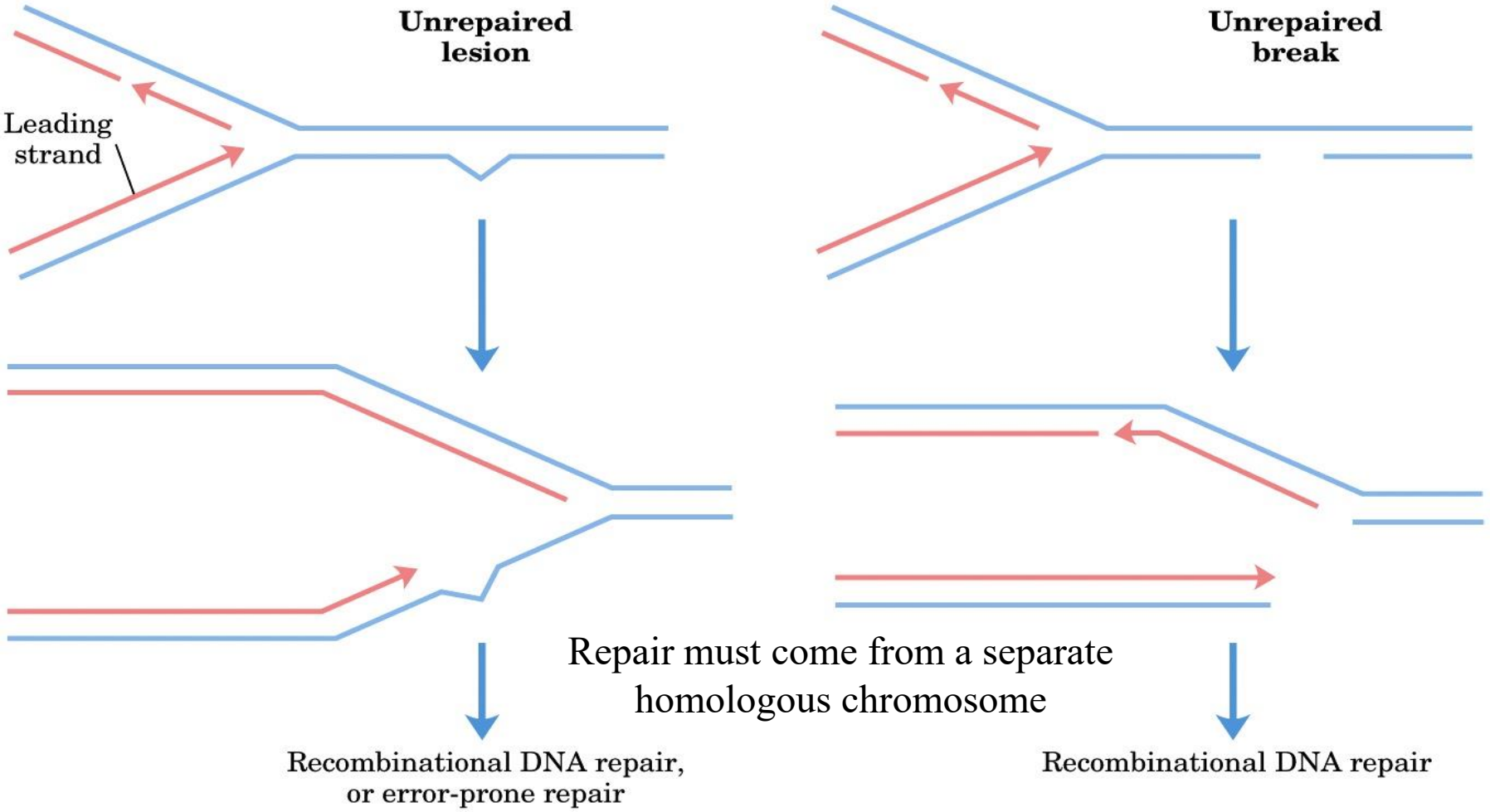
Priority given to maintain the integrity of cellular DNA.



Replication forks interacts with damage.

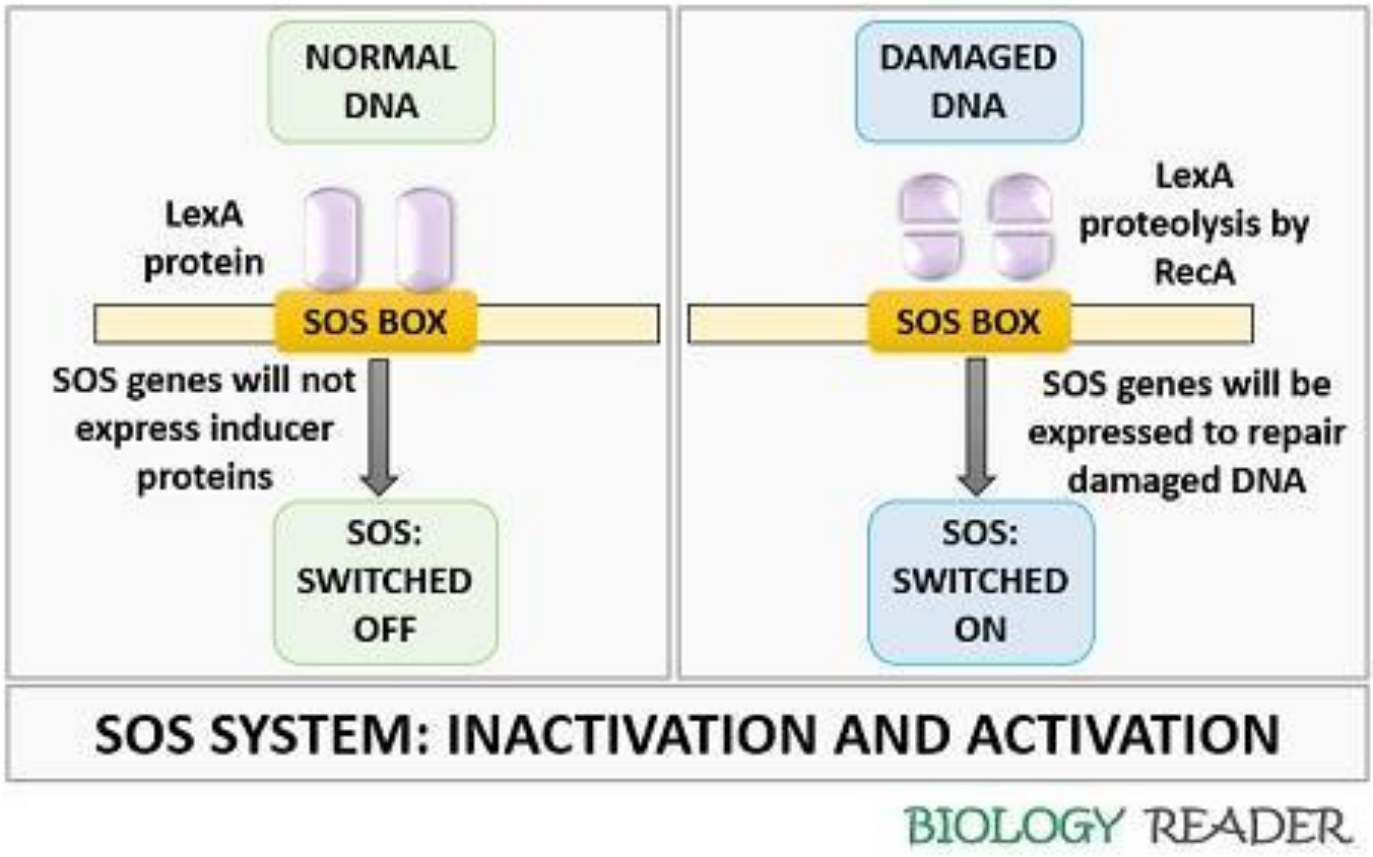
All the above repair systems work only if template is present + undamaged providing the correct genetic material to restore the wrong/damaged one.

But sometimes, due to unrepaired lesion / unrepaired break → the template is damaged/ absent (radiation, oxidative rxns).



SOS repair

- Lex A= repressor inhibits transcription of all SOS proteins.
- Rec A coprotease activity autocleavage of Lex A repressor.



Recombinational DNA repair.

Error prone translesion DNA synthesis (TLS).

Error prone = mutations often occur. Part of SOS response, SOS proteins:

DNA pol V replicate past / trans lesions that normally would block replication.

Proper base pairing is nearly impossible→ **inaccurate repair + high mutation rate.**

Lack proofreading exonuclease activity →Fidelity reduced,

DNA Recombination

DNA recombination refers to the process that a DNA segment moves from one DNA molecule to another.

1) Homologous / general recombination/ DNA cross over:

It occurs between two homologous DNA molecules.

2) Site-specific recombination:

It occurs at a specific particular DNA sequence present in both non-homologous DNA molecules.

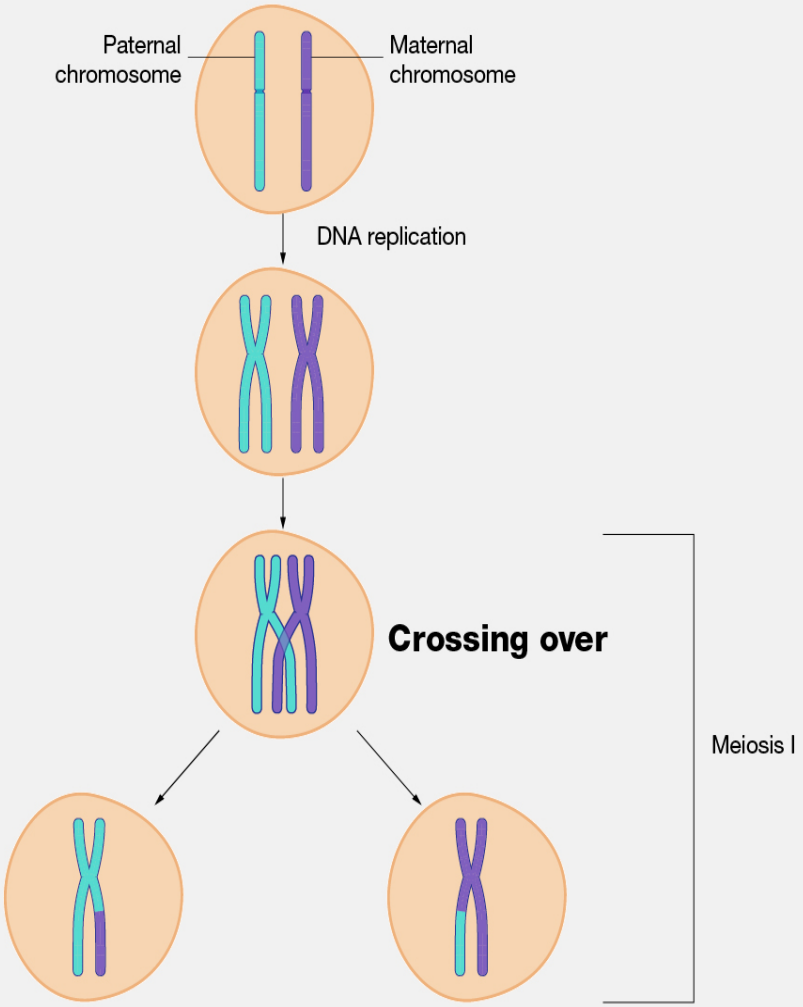
3) Transpositional recombination:

A mobile element / short segment is inserted into a target DNA.

Significance of Recombination

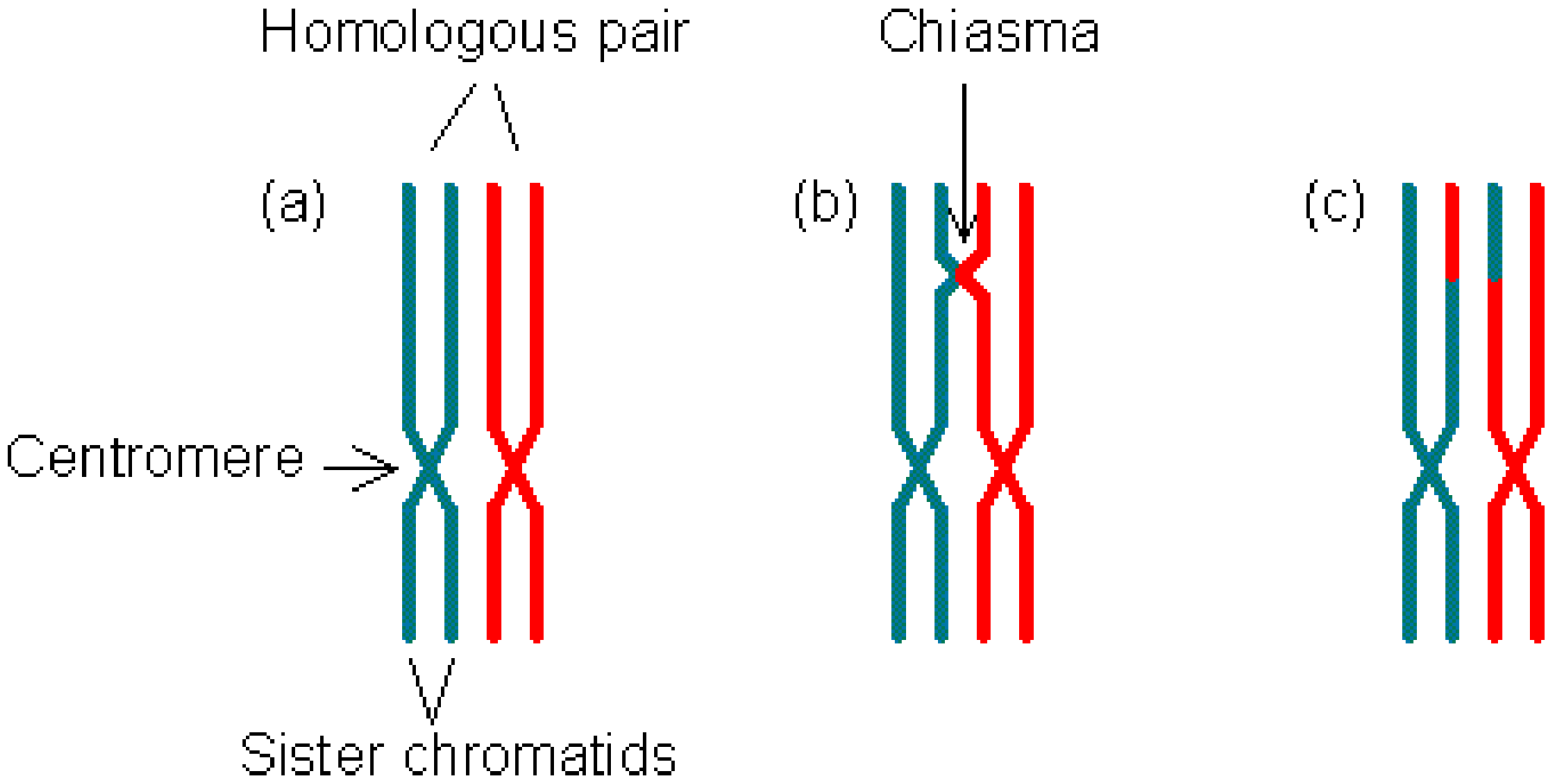
- DNA repair Recombinational repair system.
- Genetic diversity

Recombination occurs during Meiosis



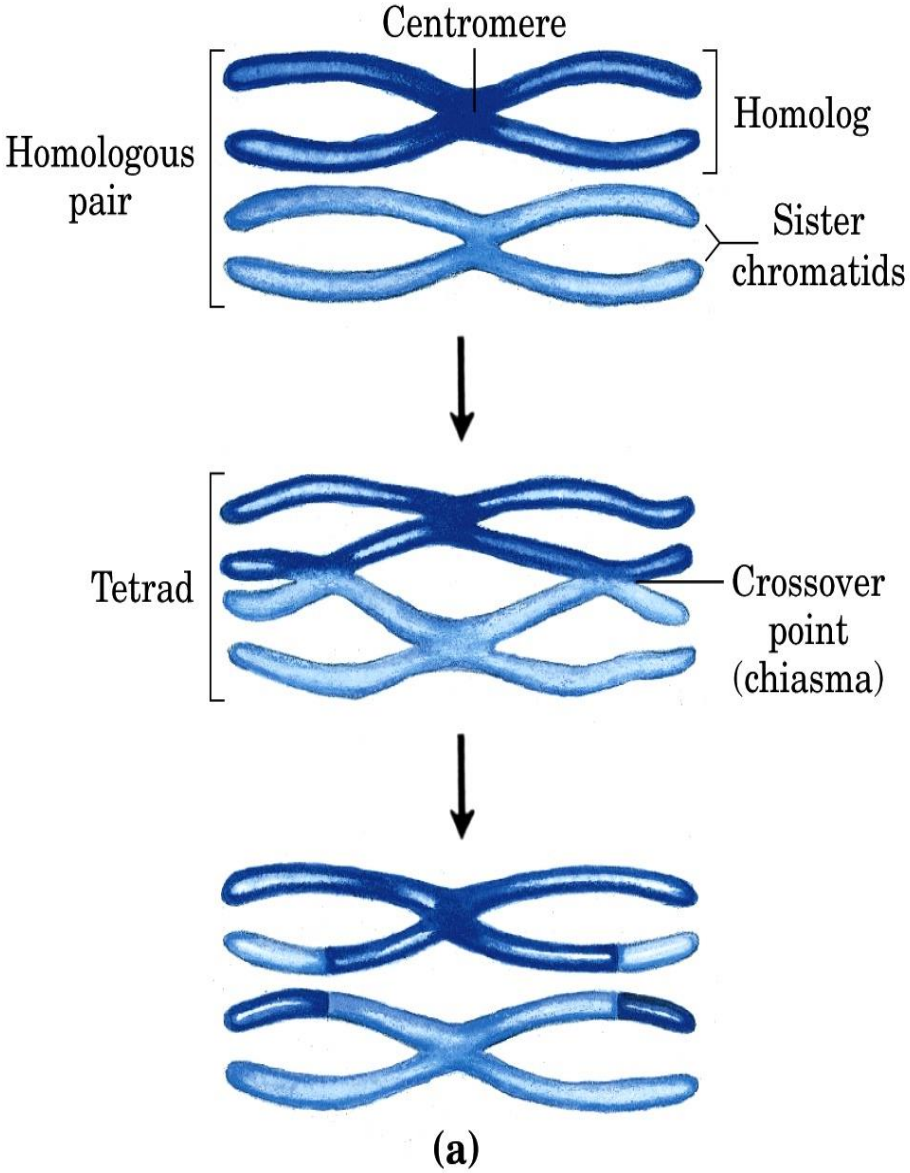
1) Homologous recombination/ DNA crossover.

Two homologous pairs of sister chromatids align side by side. **(b)** The two homologs are connected at a certain point called **chiasma**. **(c)** The two homologs exchange the DNA segment from the chiasma to the end of chromosomes.

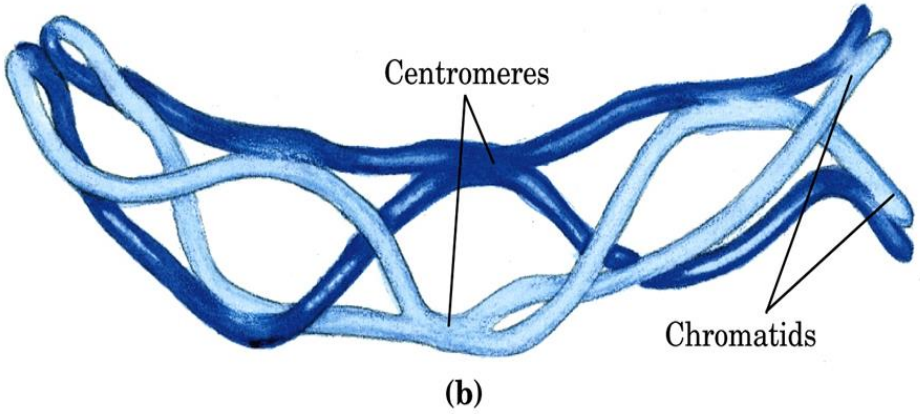


Crossing Over:

Produces exchange of genetic material.



Many points of joining chiasmata





ds Break Repair System:

Most DNA damage repaired BER/NER but replication fork encounters breaks.

Two homologous chromosomes

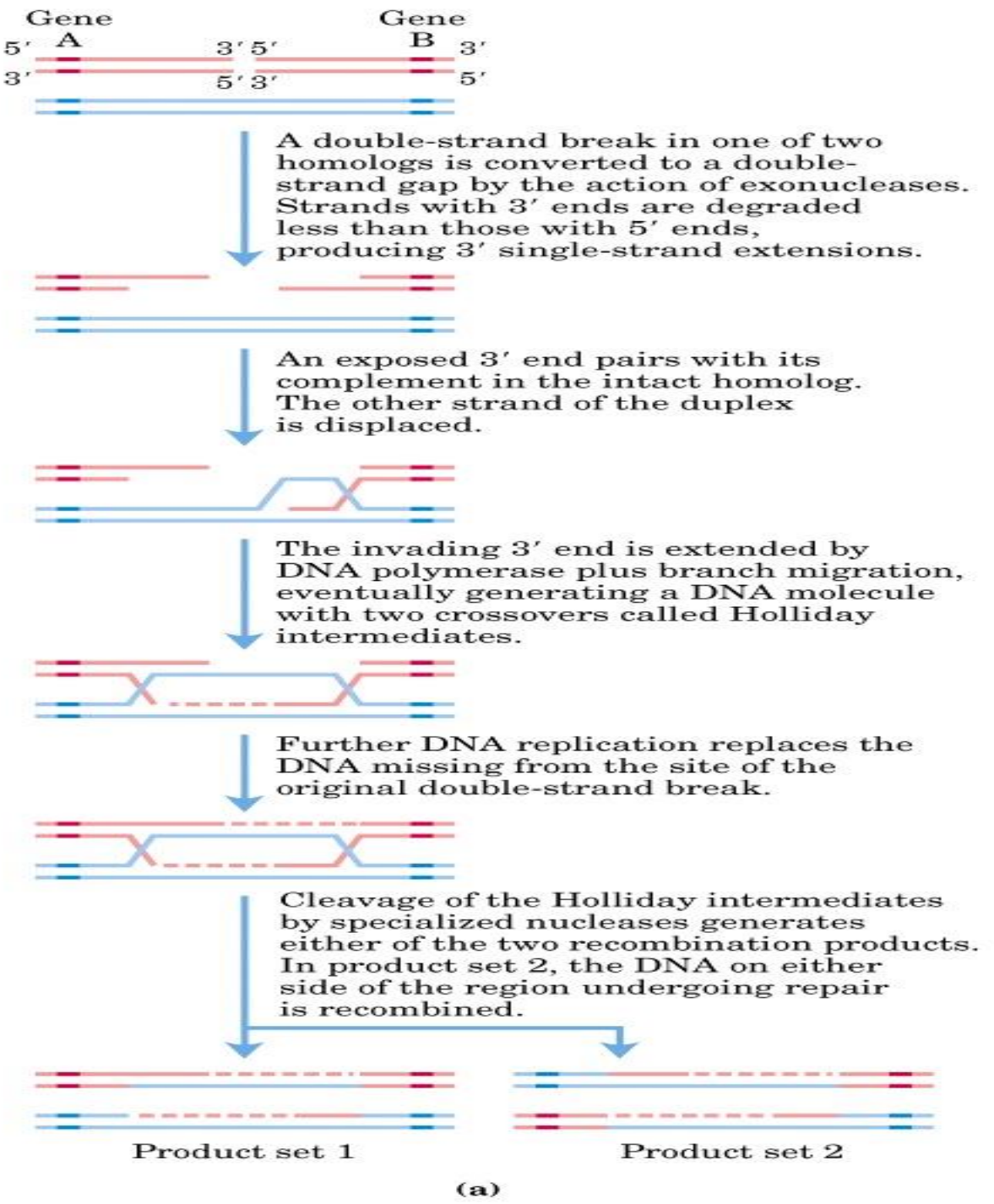
1- break→gap by exonuclease (more at 5' end)→3'ss extensions.

2- extension invades, pairs with its complementary strand in intact homolog.

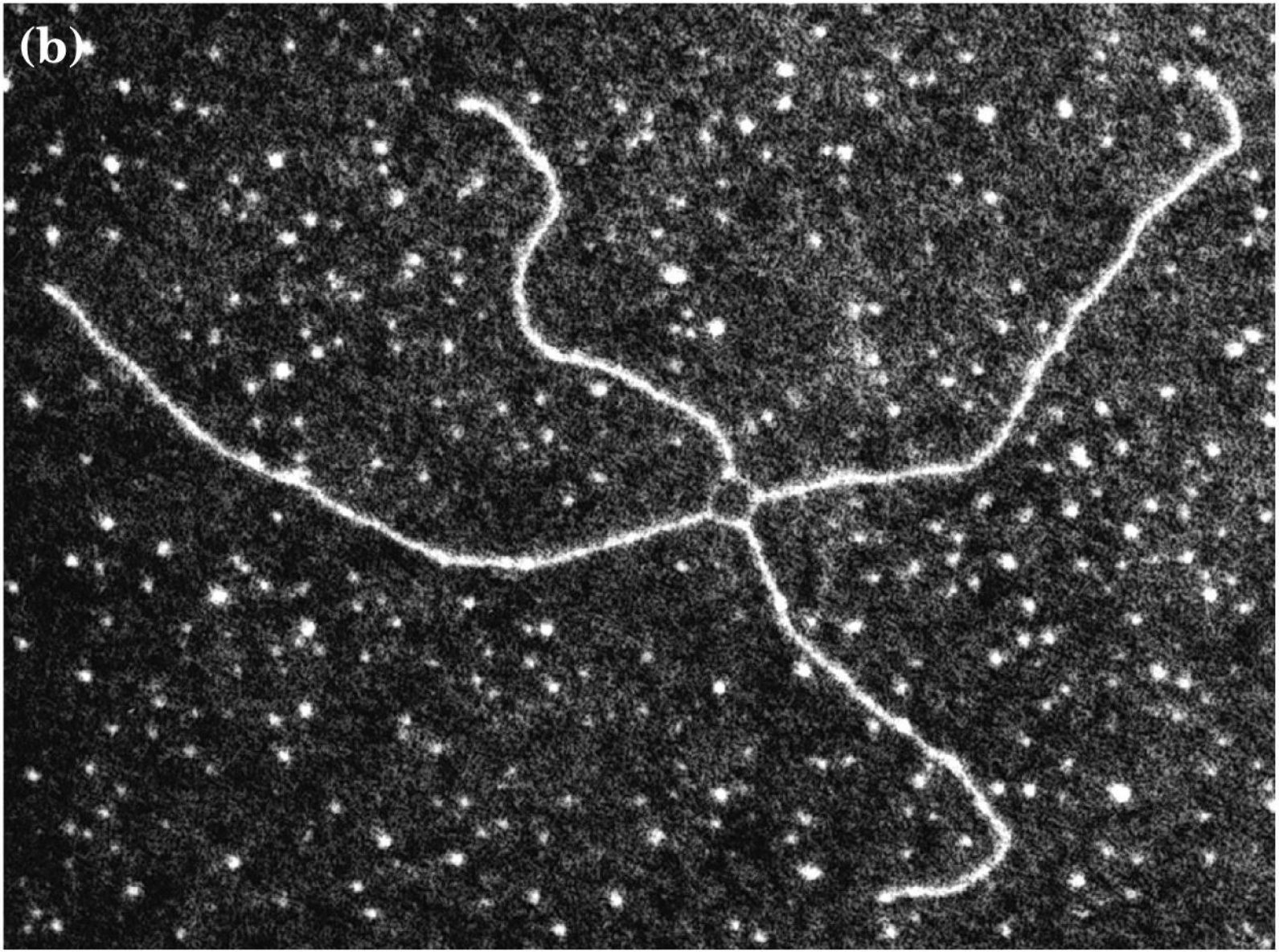
3- extended by DNA pol + branch migration.

4- two Holliday junctions.

5- specialized nucleases cleave Holiday junctions.



Holliday intermediate/ crossover junction.



- Homologous repair may vary in many details from one species to another, but most the steps are common.

- Both cleavage products are seen in vivo in eukaryotes and prokaryotes.
- Contribute to genomic diversity→
- Two homologous chromosomes are not identical→ differ slightly in different alleles.

- One contains the allele of hemoglobin A the other hemoglobin S (sickle cell anemia). A to T =glutamate to valine

- The difference one bp among millions!!!!

- Most DNA damage repaired by BER/NER but replication fork - in its journey from origin to terminus - encounters DNA ds breaks / lesions.

- DNA pol III can't continue → **Recombinational DNA repair**
  Origin-independent restart of replication:
  Complex of 7 proteins (priA, C, DnaB, C, G, T), **DNA pol II**.

- Repair of stalled/ blocked replication fork :Transition from replication→ recombinational repair → replication.

Site specific recombination:

Recombinase/integrase + recombination site (DNA segment 20-200bp)

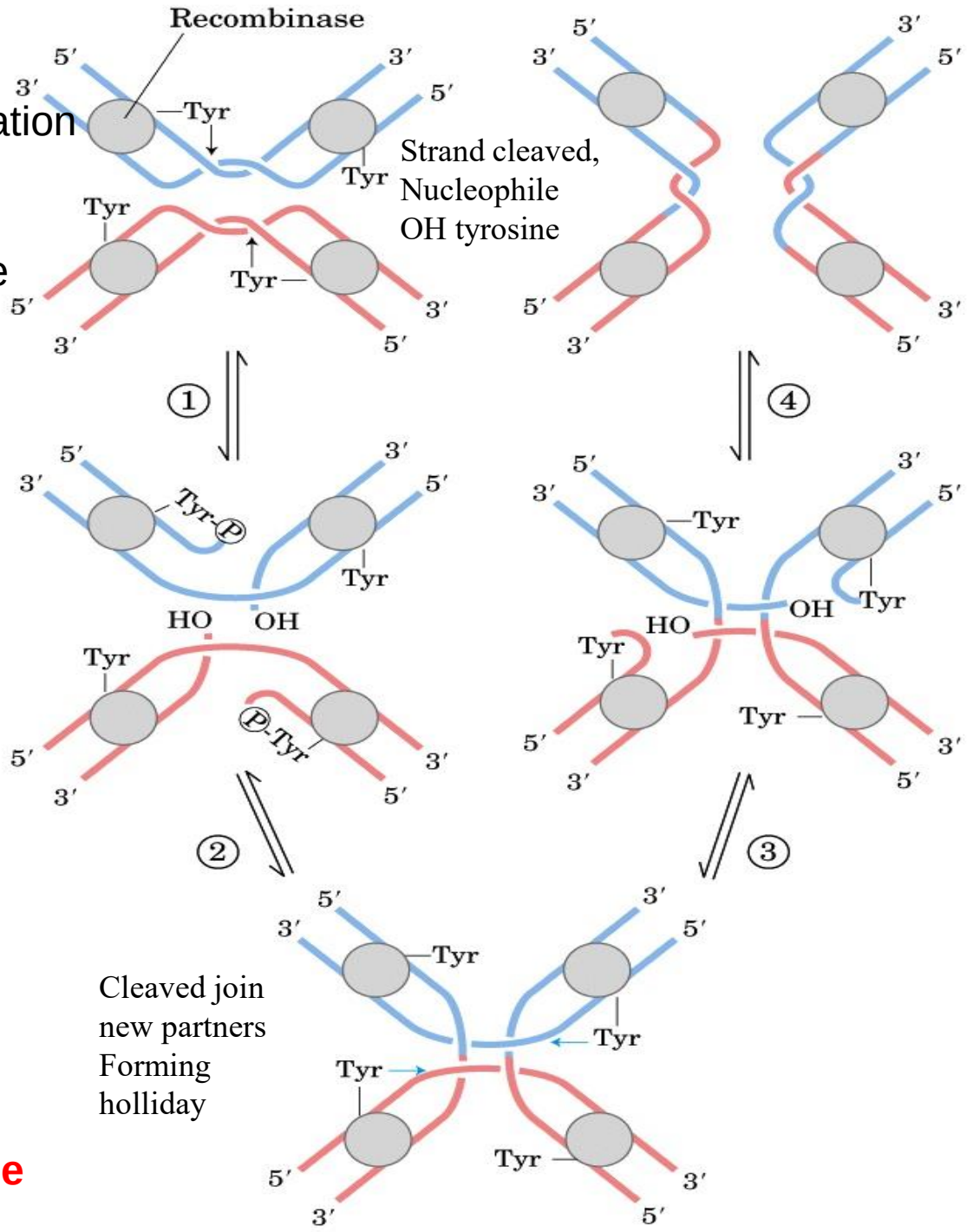
Site specific recombinase / integrase tetramer of identical units.

One recombinase recognize+bind each of the 2 recombination sites through phosphotyrosine.

Transient protein-DNA covalent linkage preserve phosphodiester bond by phosphoserine/ phosphotyrosine bond.

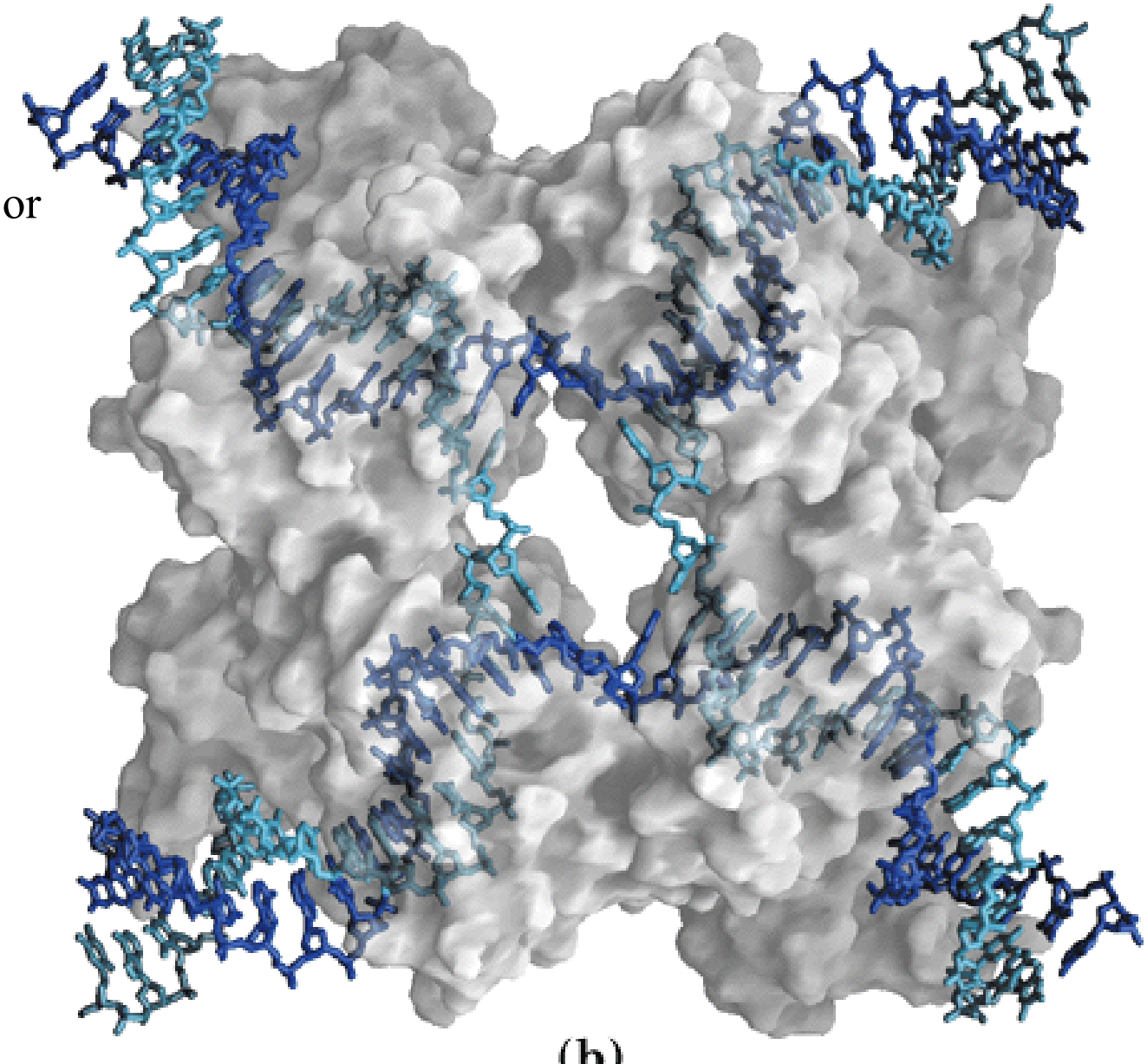
Recombinase=

site specific endonuclease + ligase



Recombinase bound to Holliday intermediate:

Strand breaking and rejoining rxns do not require DNA synthesis or degradation, i.e. no enzymes / cofactors needed.



3) Transpositional recombination:

a process in which a **mobile element** is inserted into a target DNA.

Transposable Genetic Elements:

Simple transposons:

contains only the sequence required for transposition. The genes for transposase.

Complex transposons:

contains one/ more genes in addition to those needed for transposition.

e.g. confer resistance to antibiotics.

Mechanisms of transpositional recombination

A) Class I = retrotransposons. The retrotransposon in donor DNA is first transcribed into RNA and then reverse-transcribed into DNA, which is inserted into a target DNA by the same recombination mechanism as the DNA intermediates of transposons.

B) Class II transposons. The transposon in donor DNA is cut by a special enzyme and then inserted into a target DNA. enzyme = transposase has both nuclease and ligase activities.

